03/10/2026 ITA0307 - Mobile Computing Y. Vamsi Babu

CO1 - AT2

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CSE

① 45 MHz = 45,000 Hz

Full-duplex needs 2 x 25 = 50kHz

channels = 45,000/50 = 900

∴ 900 full-duplex channels.

② channels/cells = 550/7

= 80

∴ 80 channels/cell.

③ Total Voice channels = 100 X 8

= 800

∴ 800 simultaneous channels.

④ $10 \log 10(80/4) = 10 \log 10(20) = 13.01$ dB

∴ 13.01 dB

⑤ $C/I = -60 - (-92) = 32$ dB

∴ 32 dB

⑥ Total channels = 42,000/300 = 140

Per cell = 140/7 = 20

∴ 20 channels/cell

(7) Power scales as $1/d$

Doubling distance gives $64/2 = 64/16 = 4\,W$

$\therefore$ **4 W**

(8) $P = d^2$

If distance doubles : scale $= 2^2 = 4$

$\therefore$ 4x transmit power.

(9) $4 \times 15 = 60$ channels

$\therefore$ 60 channels.

(10) $V = 100/3.6 = 27.78\ m/s$

$ID = Vf/c = (27.78 \times 900 \times 10^6) 3 \times 10) = 83.3\,Hz$

$\therefore$ **83.3Hz**

(11) $\frac{5}{8} = 0.625$ ms

$\therefore$ **0.625 ms**

(12) $24{,}000/200 = 120$

$\therefore$ 120 carriers

(13) $8\,W = 8000\,mW$

$10\log10(8000) = 39.03\,dBm$

Recieved $= 39.03 - 33 = 6.03\,dBm$

$\therefore$ **6.03 dBm**

(14) $FSPL(dB) = 32.44 + 20\log 10(fMHz) + 20\log 10(dKm)$

$= 32.44 + 2.0\log 10(1900) + 20\log 10(2.5) = 105.98\ dB$

$\therefore = 106\ dB$

(15) $6 \times 250 = 1{,}500$ calls

$\therefore$ 1,5000 simultaneous calls

(16) 3 min = 180 s

Active calls = $4000 \times 180 = 720{,}000$

$\therefore$ 720,000 active calls.

(17) $150 \times 25KHz = 3750KHz = 3.75\ MHz$

$\therefore 3.75\ MHz$

(18) 30 bands Create 29 gaps between them.

$29 \times 40KHz = 1160\ KHz = 1.16\ MHz$.

$\therefore 1.16\ MHz$

(19) Diameter $= 2 \times 1.5 = 3\ KM$

Handoff/hour $= 90/3 = 30$

$\therefore$ 30 handoffs / hour

(20) ~~Diameter = 2x+~~

$8 \times 40 \times 250 = 800$ users

$\therefore 80{,}000$ users.

(21) $R = 0.4$ km. Hexagonal area $= (3\sqrt{3}/2)R^2$

$= 0.4157 \text{ km}^2$

Cells $= 400/0.4157 = 962$, so round up to 963 cells to cover the region.

(22) $6,000/30 = 200$ channels

$\therefore$ 200 channels.

(23) Traffic $= 1200 \times 4 \times (2.5/60) = 200$ Erlangs.

$\therefore$ 200 Erlangs.

(24) $40,000/400 = 100$ users

$\therefore$ 100 simultaneous users.

(25) $1200 \times 600 = 720,00$ users.

$\therefore$ 100 simultaneous users.

(26) Rounds $= 200/4 = 50$ rounds

$\therefore$ 50 paging rounds

(27) offered traffic $= 40,000 \times 3 = 120,000$ Erlangs equivalent calls.

Capacity $= 100,000$ Blocked $= 20,000$

Blocked fraction $= 20,000/120,000 \times 100 = 16.67\%$

$\therefore 16.67\%$

(28) $600,000 \times 0.16 = 96,000$

$\therefore$ 96,000 subscribers.

12,000 - 9,000 = 3,000 calls/s

(29) ∴ 3,000 calls/s spare

(30) Total channels = 48,000/200 = 240

Per cell = 240/4 = 60 channels.

∴ 60 channels/cell.

(31) 30,000/200 = 150 channels

∴ 150 channels.

(32) 2500 × 0.99 = 2475 successful handoffs

∴ 2,475

(33) 12 W = 12,000 mW

10 log 10 (12000) = 40.79 dBm.

Recieved = 40.79 - 45 = -4.21 dBm.

∴ -4.21 dBm

(34) 7 × 900 = 6,300 Users

∴ 6,3000 Simoltaneous Users.

(35) 4.8/8 = 0.6 MS

∴ 0.6 MS

(36) V = 150/3.6 = 41.67 M/s

fD = Vf/c = (41.67 × 1.8 × 10)(3 × 10) = 250 Hz

∴ 250 Hz

37. Offered load = 70,000 × 4 = 280,000

Blocked = 280,000 − 250,000 = 30,000

Blocked share = 30,000 / 280,000 × 100 = 10.71%

∴ = 10.71%

38. 9 × 600 = 5,400 Users

∴ 5,400 Users.

39. 60,000 × 0.004 = 240 calls/s

= 240 calls/s

40. Path loss = 43 − (−87) = 130 dB.

∴ 130 dB.

41. 150 × 2.5 = 375 km^3

∴ 375 km^2

42. Total channels = 48,000 / 200 = 240

Per cell = 240/6 = 40 channels

∴ 40 channels/cell

43. 150,000 / 6,000 = 25 cycles

25 paging cycles.

(44) $8/3 = 2.67$ call.

* This corresponds to roughly 3 handoffs over the

∴ 3 handoffs.

(45) $SNR = -85 - (-115) = 30\,dB$

∴ 30 dB

(46) $250 \times 150\ Kbps = 37,500\ Kbps = 37.5\ Mbps$

∴ 37.5 Mbps

(47) $Traffic = 1500 \times 4 \times (2.5/60) = 250$ Erlangs

∴ 250 Erlangs.

(48) $250 \times 12KHz = 3000KHz = 3\ MHz$

∴ 3 MHz

(49) $1500 \times 600\ Kbps = 900,000\ Kbps = 900\ Mbps$

∴ 900 Mbps

(50) $Area/tower = \pi^2(4^2) = 50.27\ km^2$

$Towers = 350/50.27 = 6.96$, so round up

∴ 7 towers.